

Bayer Leverkusen 23-24

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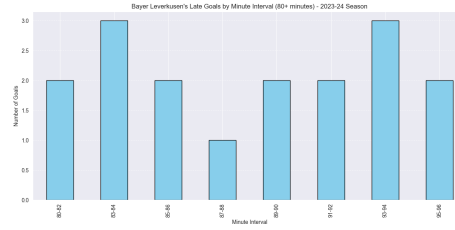


Figure 1: Late Goals

1 Introduction

In the 2023/2024 season, Bayer Leverkusen has emerged as a dominant force in German football, with their success largely attributed to a well-balanced and tactically astute squad. At the heart of their impressive campaign are two key players: Florian Wirtz and Granit Xhaka. This report delves into the pivotal roles these players have assumed within Bayer Leverkusen's system, analyzing their contributions through a comprehensive examination of statistical data and performance metrics.

Wirtz, a young attacking midfielder, has established himself as one of the Bundesliga's most exciting talents. His creative flair and ability to influence games in the final third have been instrumental in Leverkusen's attacking prowess. Xhaka, on the other hand, has redefined his career since joining Leverkusen, becoming the linchpin of the midfield with his expansive passing range and tactical intelligence.

Through an in-depth analysis of their touch heatmaps, passing networks, defensive contributions, and key performance indicators, this report aims to shed light on how Wirtz and Xhaka complement each other and form the backbone of Leverkusen's tactical setup. We will explore their roles in ball progression, chance creation, and maintaining tactical balance, while also examining how their performances compare to other key players in the squad.

By dissecting their individual and combined impacts, we seek to understand the tactical nuances that have propelled Bayer Leverkusen to the forefront of German and European football. This analysis will not only highlight the importance of Wirtz and Xhaka to their team but also provide insights into the evolving nature of midfield play in modern football.

2 Analysis

2.1 Late Goals Analysis

The bar chart [fig:1] illustrates Bayer Leverkusen's late goal-scoring patterns during the 2023-24 season, focusing on goals scored after the 80th minute. The data reveals a consistent threat in the final stages of matches, with notable peaks in the 83-84 and 93-94 minute intervals, each showing 3 goals scored.

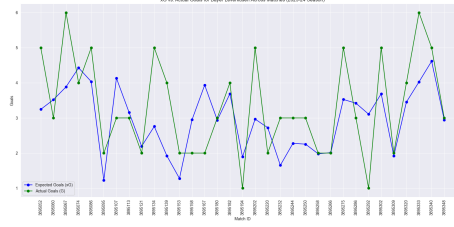


Figure 2: Xg vs Goals

Most other intervals maintain a steady output of 2 goals, with only the 87-88 minute period showing a slight dip to 1 goal. This pattern suggests strong endurance, tactical adaptability, and mental fortitude from Bayer Leverkusen, enabling them to capitalize on opportunities late in games. The team’s ability to score consistently in stoppage time, as evidenced by the 93-94 minute peak, further underscores their persistence and potential impact on match outcomes.

2.2 Expected Goals (xG) vs. Actual Goals

This line graph [fig:2] compares Bayer Leverkusen’s Expected Goals (xG) with their Actual Goals (G) across matches in the 2023-24 season. The x-axis represents individual matches by ID, while the y-axis shows the number of goals, ranging from 0 to 6. The blue line depicts xG, and the green line represents actual goals scored.

The graph reveals significant fluctuations in both xG and actual goals from match to match. Notably, there are several instances where actual goals exceed xG, indicating efficient finishing (e.g., match IDs 3959067 and 3959333). Conversely, there are matches where xG surpasses actual goals, suggesting missed opportunities (e.g., match ID 3959107).

Overall, the graph demonstrates Bayer Leverkusen’s attacking prowess, with actual goals frequently matching or exceeding xG. This pattern indicates a combination of high-quality chance creation and clinical finishing throughout the season, contributing to the team’s strong offensive performance.

2.3 Passing Network

This passing network diagram [fig:3] illustrates Bayer Leverkusen’s tactical approach and player interactions on the pitch. The visualization depicts players as nodes, with their jersey numbers displayed, while the connecting lines represent passing patterns between them. The size of each node correlates with the total number of passes made by that player, ranging from 0 to 3400 passes.

Central to the network is player 34, Granit Xhaka, whose large node size indicates their pivotal role in ball distribution. Strong connections between central players 34, 8, 12, and 4 suggest a midfield-centric passing strategy. The

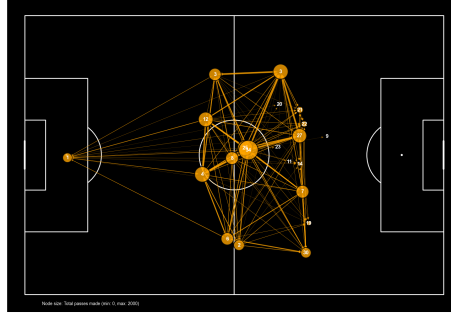


Figure 3: Passing Network

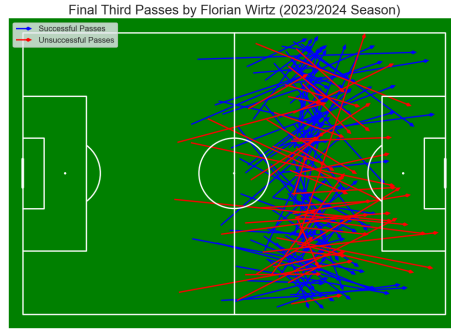


Figure 4: Passes to final third by Wirtz

goalkeeper, 1, shows multiple connections to outfield players, implying active involvement in the team’s build-up play. Wide players 3, 22, and 30 are positioned near the touchlines, providing width to the formation, while forward players 27 and 19 are placed higher up the pitch with clear links to the midfielders.

This network visualization effectively showcases Bayer Leverkusen’s emphasis on central midfield play, their use of wide attacking options, and a possession-based style that incorporates the goalkeeper into the team’s overall distribution strategy. The intricate web of connections between players across different areas of the pitch reflects a fluid and versatile playing style.

2.4 Progressive Passes to Final Third

These two images [fig:4] , [fig:5] illustrate the final third passes made by Granit Xhaka and Florian Wirtz for Bayer Leverkusen during the 2023/2024 season. Both visualizations use a soccer pitch diagram to map the passes, with successful passes represented by green (for Xhaka) or blue (for Wirtz) lines, and unsuccessful passes shown in red.

Granit Xhaka’s passing map reveals a high volume of passes into the final

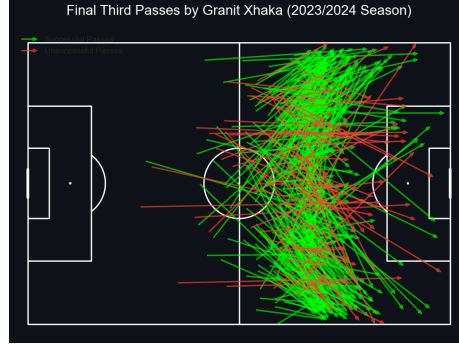


Figure 5: Passes to final third by Xhaka

third, predominantly from central and left-sided positions. The majority of his passes are successful, indicated by the dense concentration of green lines. Xhaka's passes cover a wide range of the pitch, suggesting his role as a deep-lying playmaker who can distribute the ball to various attacking positions.

Florian Wirtz's passing map shows a different pattern. His passes are concentrated more in the central and right areas of the final third, indicating his role as an advanced playmaker or attacking midfielder. Wirtz's successful passes (blue lines) outnumber his unsuccessful ones, but there's a more even distribution between successful and unsuccessful passes compared to Xhaka's map.

Both players demonstrate significant involvement in Bayer Leverkusen's attacking play, with Xhaka providing a broader range of passes from deeper positions and Wirtz focusing on more intricate, advanced passes in and around the penalty area. This combination of passing styles likely contributes to the team's diverse attacking approach and ability to create chances from different areas of the pitch.

2.5 Progressive Passes to Final Third by Xhaka to wirtz

This image [fig:6] visualizes the progressive passes from Granit Xhaka to Florian Wirtz that reach the final third of the pitch for Bayer Leverkusen during the 2023/2024 season. The diagram uses a soccer pitch layout with green arrows representing successful passes.

Xhaka made a total of 393 passes to Wirtz, with 60 of these progressing into the final third. The passes originate from various positions in the middle and defensive thirds, with a notable concentration in the central and left-central areas. The arrows terminate in diverse locations across the final third, indicating Wirtz's dynamic movement and positioning.

Xhaka demonstrates an ability to make both short and long-range passes into the final third, showcasing his versatility as a distributor. This visualization highlights Xhaka's role as a key playmaker, particularly in finding Wirtz in advanced positions. The variety in pass origins and destinations suggests a dy-

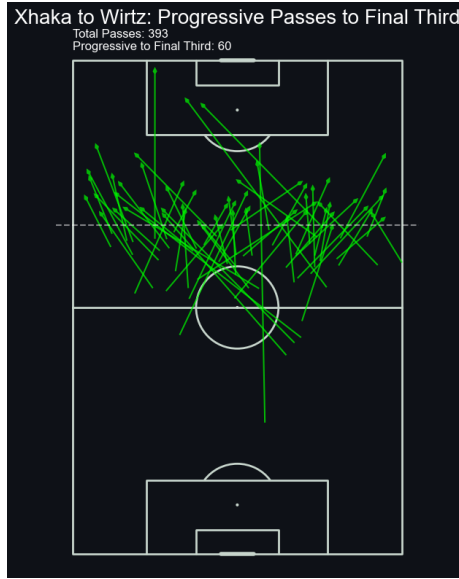


Figure 6: Progressive Passes to Final Third by Khaka to wirtz

namic partnership between the two players, contributing significantly to Bayer Leverkusen’s attacking play and ball progression strategies.

Overall, this passing map underscores the importance of the Khaka-Wirtz connection in Bayer Leverkusen’s offensive tactics, illustrating how Khaka’s vision and passing range complement Wirtz’s movement in the final third.

2.6 PPDA (Passes Per Defensive Action)

This horizontal bar chart [fig:7] illustrates the Passes Per Defensive Action (PPDA) for Bayer Leverkusen players during the 2023/2024 season. PPDA is a metric used to measure the intensity of a team’s or player’s pressing, with

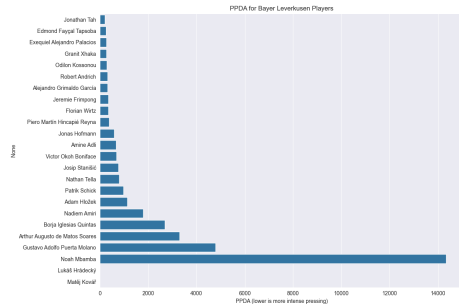


Figure 7: PPDA

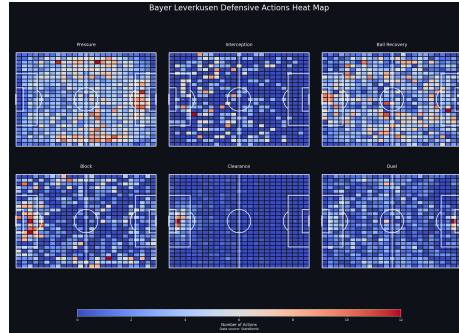


Figure 8: defensive actions heat map

lower values indicating more intense pressing.

The chart shows a wide range of PPDA values across the squad. Players like Jonathan Tah, Edmond Tapsoba, Exequiel Palacios, and Granit Xhaka have very low PPDA values, suggesting they are among the most active and intense pressers in the team. This aligns with their roles as defenders and midfielders who are often involved in winning back possession.

In contrast, players such as Noah Mbamba, Gustavo Puerta, and Arthur Soares have significantly higher PPDA values, which could indicate they play in positions or roles that involve less pressing, or they may have had limited playing time.

Interestingly, some attacking players like Florian Wirtz and Amine Adli show relatively low PPDA values, suggesting they contribute significantly to the team's pressing from the front.

The variation in PPDA across the squad reflects Bayer Leverkusen's tactical approach, which likely involves intense pressing from certain key players while others may focus on different aspects of the game. This data provides insight into the team's defensive strategy and the specific roles of individual players within that system.

2.7 Defensive Actions Heat Map

This image [fig:8] presents a comprehensive heat map of Bayer Leverkusen's defensive actions during the 2023/2024 season, divided into six categories: Pressure, Interception, Ball Recovery, Block, Clearance, and Duel. Each subplot represents a soccer pitch, with color intensity indicating the frequency of actions in specific areas.

Pressure: Shows a high concentration in the middle third and wide areas, suggesting an aggressive pressing strategy across the pitch, with particular emphasis on wide areas and the opponent's half.

Interception: Displays a more even distribution across the field, with slightly higher intensity in central areas, indicating the team's ability to read and disrupt opponent's passes throughout the pitch.

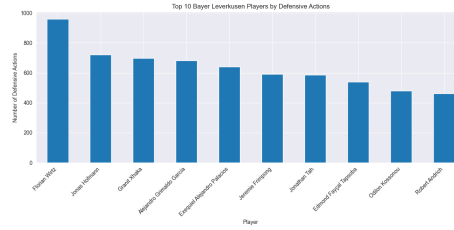


Figure 9: Top Defensive Players

Ball Recovery: Exhibits a pattern similar to interceptions but with higher intensity in the defensive third, showcasing the team’s proficiency in regaining possession, especially in their own half.

Block: Concentrated heavily in the defensive third, particularly in front of the goal and in wide defensive areas, highlighting the team’s commitment to protecting their goal.

Clearance: Primarily focused in the defensive third and penalty area, as expected, demonstrating the team’s ability to alleviate pressure in dangerous areas.

Duel: Shows a fairly even distribution across the pitch, with slightly higher intensity in wide areas and the defensive third, indicating the team’s engagement in one-on-one contests throughout the field.

Overall, this heat map reveals Bayer Leverkusen’s well-rounded defensive approach, combining high pressing, strategic interceptions, and solid defensive positioning. The team appears to defend actively across all areas of the pitch, with a particular emphasis on wide areas and their own defensive third. This comprehensive defensive strategy likely contributes significantly to their overall tactical approach and success in the season.

2.8 Top Defensive Players

This bar chart [fig:9] displays the top 10 Bayer Leverkusen players ranked by their number of defensive actions during the 2023/2024 season. The y-axis represents the number of defensive actions, while the x-axis lists the players’ names. Florian Wirtz leads the chart with nearly 1000 defensive actions, which is notably high for a player typically associated with an attacking role. This suggests Wirtz’s significant contribution to the team’s pressing and defensive efforts.

Following Wirtz, Jonas Hofmann and Granit Xhaka occupy the second and third positions respectively, each with over 700 defensive actions. Their high rankings align with their midfield roles, which often involve both attacking and defensive responsibilities.

Defenders like Alejandro Grimaldo Garcia and Jonathan Tah are also present in the top 10, as expected given their primary defensive duties. The presence of Exequiel Palacios and Jeremie Frimpong in the middle of this ranking indicates

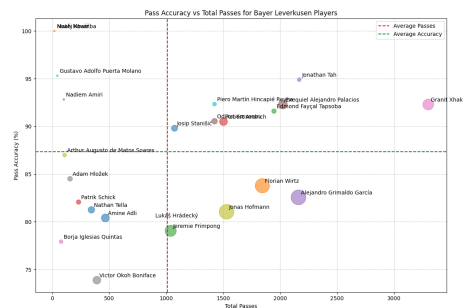


Figure 10: Pass Accuracy vs. Total Passes

their active involvement in the team's defensive strategy.

The chart reveals that Bayer Leverkusen's defensive approach is not solely reliant on traditional defenders, but involves significant contributions from midfielders and even attacking players. This data supports the notion of a team-wide commitment to defensive duties and potentially indicates a high-pressing tactical style employed by the team.

The relatively even distribution of defensive actions among these top players (all performing over 400 actions) suggests a balanced defensive workload across different positions in the team.

2.9 Pass Accuracy vs. Total Passes

This scatter plot [fig:10] illustrates the relationship between pass accuracy and total passes for Bayer Leverkusen players in the 2023/2024 season. The x-axis shows the total number of passes, while the y-axis represents pass accuracy percentage. Bubble sizes correspond to the player's total passes, with larger bubbles indicating higher pass volumes.

Granit Xhaka emerges as a standout performer, positioned in the top right corner with over 3000 total passes and approximately 93 percent accuracy. This highlights his crucial role in the team's ball distribution and possession play. Florian Wirtz and Alejandro Grimaldo García also demonstrate high pass volumes (around 2000-2500) but with slightly lower accuracy (83-84percent) compared to some teammates, possibly due to their more advanced and creative roles.

Players like Jonathan Tah, Exequiel Palacios, and Edmond Tapsoba exhibit a balance of high pass accuracy (above 90percent) and substantial pass numbers (1500-2500 range), underscoring their importance in building play from the back and midfield. Interestingly, some players such as Noah Mbamba and Nadiem Amiri show very high pass accuracy (near 100percent) but with lower total passes, which might indicate limited playing time or specialized roles within the team. The plot includes average lines for pass accuracy (green dashed, around 87.5percent) and total passes (red dashed, about 1000), providing context for individual performances. Most regular starters cluster in the upper right quad-

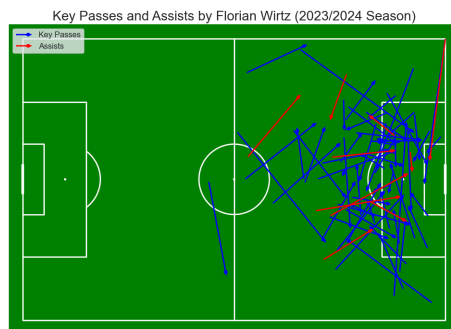


Figure 11: Key Passes Wirtz

rant, combining high pass volumes with good accuracy, which aligns with Bayer Leverkusen’s possession-based playing style.

This visualization effectively captures the passing proficiency of Bayer Leverkusen’s squad, revealing key contributors to the team’s ball circulation and highlighting the varied roles and responsibilities within the team’s tactical setup.

2.10 Key passes and assists by Wirtz

This image [fig:11] displays a soccer pitch diagram illustrating Florian Wirtz’s key passes and assists for Bayer Leverkusen during the 2023/2024 season. Blue arrows represent key passes, while red arrows indicate assists.

The visualization reveals Wirtz’s significant creative influence in the attacking third of the pitch. There’s a dense concentration of both key passes and assists in and around the penalty area, showcasing his ability to create chances in high-value areas. Many of his key passes originate from central positions just outside the box, indicating his role as a central playmaker. Wirtz also demonstrates versatility in his passing range, with some longer passes from deeper positions and the flanks. His assists are particularly concentrated within and around the penalty area, highlighting his effectiveness in creating high-quality scoring opportunities for teammates.

The spread of passes across the width of the pitch suggests Wirtz’s ability to influence play from various positions, though there’s a clear preference for central and right-sided areas. This aligns with his likely role as an attacking midfielder or inside forward.

Overall, the diagram underscores Wirtz’s crucial role in Bayer Leverkusen’s attacking play, showcasing his vision, passing accuracy, and ability to create scoring chances from diverse positions on the field.

2.11 Final 3rd Progressions

This horizontal bar chart [fig:12] illustrates the total final third progressions for Bayer Leverkusen players during the 2023/2024 season. The players are ranked

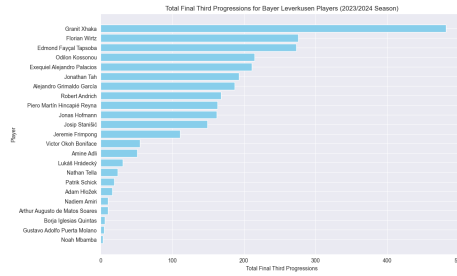


Figure 12: Final Third Progressions

in descending order based on their number of progressions into the final third of the pitch.

Granit Xhaka leads the chart with nearly 500 final third progressions, significantly ahead of his teammates. This underscores Xhaka’s crucial role in transitioning the ball from midfield to attacking positions, highlighting his importance in Bayer Leverkusen’s build-up play.

Florian Wirtz and Edmond Fayçal Tapsoba follow closely, each with over 300 progressions. Wirtz’s high number reflects his role as an attacking midfielder, often receiving the ball in advanced positions, while Tapsoba’s performance is notable for a defender, indicating his involvement in initiating attacks from the back.

Players like Odilon Kossounou, Exequiel Alejandro Palacios, and Jonathan Tah also show significant contributions, with each making over 200 progressions. This demonstrates the team’s strategy of progressing the ball from various positions on the field.

The chart reveals a gradual decrease in progressions moving down the list, with players like Jeremie Frimpong, Victor Okoh Boniface, and Amine Adli still making notable contributions despite fewer progressions.

Players at the bottom of the chart, such as Noah Mbamba and Gustavo Adolfo Puerta Molano, have very few progressions, which might be due to limited playing time or defensive roles that don’t typically involve progressing the ball into the final third.

Overall, this visualization highlights Bayer Leverkusen’s approach to ball progression, with contributions from various positions, emphasizing the team’s collective effort in moving the ball into dangerous areas of the pitch.

2.12 Final Third Carries by Wirtz

The visualization [fig:13] reveals Wirtz’s significant involvement in progressing the ball into and within the final third. There’s a high concentration of carries starting just before the final third and continuing into more advanced positions. This demonstrates Wirtz’s ability to transition the team from midfield into attacking areas through his dribbling skills.

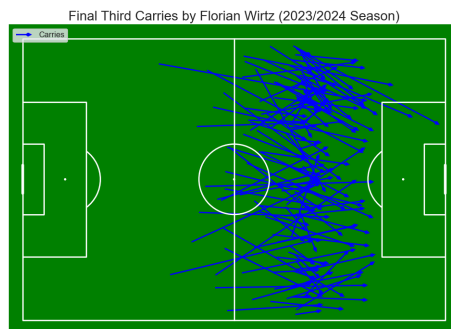


Figure 13: Final 3rd carries

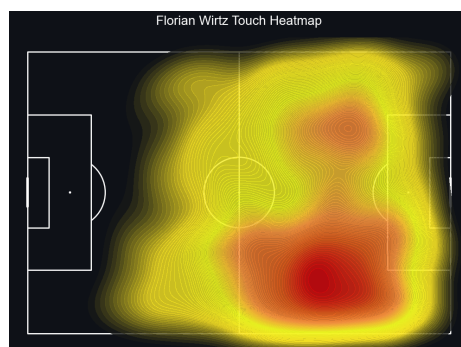


Figure 14: Florian Wirtz Heatmap

Many of the carries originate in central areas, but there's also a notable spread to both the left and right sides of the pitch. This suggests Wirtz's versatility in operating across the width of the field, though there appears to be a slight preference for the right side.

The carries vary in length, with some covering significant distances while others are shorter, more intricate movements in tight spaces near the opponent's penalty area. This highlights Wirtz's adaptability in navigating different defensive scenarios.

Several carries penetrate deep into the penalty area, indicating Wirtz's willingness and ability to take on defenders in dangerous positions close to goal.

Overall, this diagram underscores Wirtz's crucial role in Bayer Leverkusen's attacking play, showcasing his dribbling prowess, spatial awareness, and ability to progress the ball into threatening positions across the final third of the pitch.

2.13 Touch Heatmaps

These two touch heatmaps [fig:14] , [fig:15] illustrate the on-pitch activity of Florian Wirtz and Granit Xhaka for Bayer Leverkusen during the 2023/2024

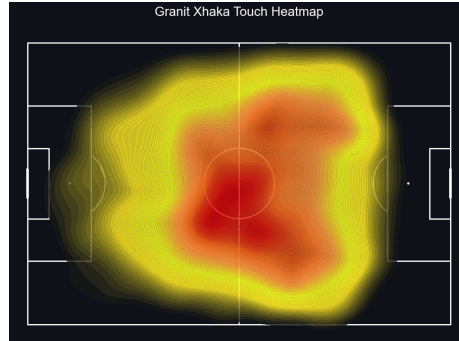


Figure 15: Granit Xhaka Heatmap

season. The color intensity, ranging from yellow (less frequent) to red (most frequent), indicates the concentration of touches in different areas of the field.

Florian Wirtz's heatmap reveals his primary influence in the attacking third, with the highest concentration of touches in the right-central area. His activity spreads across the width of the pitch in the final third, demonstrating his versatility in attack. Wirtz shows some involvement in deeper areas, particularly on the right side, but has limited activity in defensive zones. This pattern suggests Wirtz operates mainly as an attacking midfielder or right-sided forward, with the freedom to roam and create opportunities across the attacking third.

In contrast, Granit Xhaka's heatmap displays intense activity in the central midfield area, especially just inside his own half. His influence extends broadly across the pitch width in both halves, indicating his pivotal role in the team's overall play. Xhaka exhibits significant involvement in the defensive third, particularly on the left side, while also showing some presence in attacking areas, though less intensely than in deeper positions. This heatmap portrays Xhaka as a central midfielder with substantial defensive responsibilities who also contributes significantly to the team's build-up play.

Comparing the two heatmaps highlights the complementary roles of Wirtz and Xhaka in Bayer Leverkusen's system. Wirtz's focus on the attacking third contrasts with Xhaka's more comprehensive field coverage, reflecting their distinct positional roles. Xhaka's substantial defensive activity and broader pitch coverage underscore his box-to-box midfielder role, while Wirtz's concentrated presence in advanced areas emphasizes his attacking specialization.

These visualizations effectively demonstrate how Wirtz and Xhaka work in tandem within Bayer Leverkusen's tactical setup. Xhaka provides a solid mid-field foundation and distribution from deep positions, enabling Wirtz to focus on creating and finishing chances in more advanced areas. This combination likely contributes significantly to the team's overall tactical approach and success on the field.

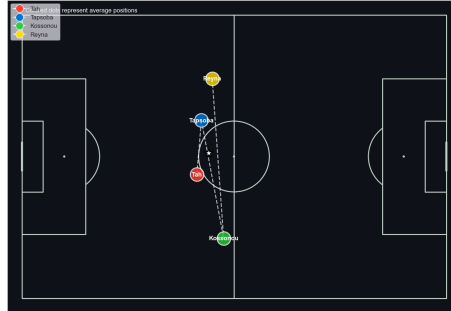


Figure 16: Average Positions across 23-24 season

2.14 Bayer Leverkusen's Defensive Core: Spatial Analysis of Key Players

This visualization [fig:16] illustrates the defensive structure and positioning of key players in Bayer Leverkusen's setup during the 2023/2024 season. The image displays a soccer pitch diagram with colored dots representing the average positions of four crucial defensive players: Tah, Tapsoba, Kossounou, and Reyna.

Tah (red) and Tapsoba (blue) are positioned as central defenders, with Tah slightly to the left and Tapsoba to the right. This arrangement forms the core of Leverkusen's defensive line. Kossounou (green) is situated as a defensive midfielder, positioned just ahead of the center-backs, serving as a link between defense and midfield. Reyna (yellow) takes up a more advanced position on the pitch, likely functioning in an attacking midfield or forward role.

The dotted lines connecting these players indicate potential defensive shapes and passing connections. This formation suggests a flexible system, possibly alternating between a back three and a back four, with Kossounou's position being key to this versatility. His ability to drop between or in front of the center-backs provides additional defensive cover and facilitates the initiation of attacks from deep. This defensive setup reveals Bayer Leverkusen's tactical approach to maintaining defensive solidity while also preparing for quick transitions into attack. The positioning of Tah and Tapsoba as the central defensive pair provides a strong foundation, while Kossounou's role adds depth and flexibility to the defensive structure. Reyna's advanced position hints at the team's commitment to maintaining an attacking threat even when in a defensive stance.

The visualization effectively demonstrates how Leverkusen balances defensive responsibilities with attacking intentions through clever player positioning. It showcases the team's spatial organization and highlights the importance of positional play in their overall tactical strategy. This defensive arrangement likely plays a crucial role in the team's ability to control games and transition smoothly between defensive and offensive phases of play.

3 Conclusion

In conclusion, this comprehensive analysis of Bayer Leverkusen’s 2023/2024 season reveals a tactically sophisticated team built around the complementary skills of key players. The data-driven insights gleaned from touch heatmaps, passing networks, and defensive positioning charts illuminate the team’s strategic approach and individual player contributions.

Florian Wirtz emerges as a creative fulcrum in the attacking third, his heatmap showcasing his influence across the width of the pitch, particularly in right-central areas. His ability to progress the ball into dangerous areas, coupled with his key passes and assists, underlines his importance to Leverkusen’s offensive play. In contrast, Granit Xhaka’s heatmap depicts a player central to both defensive stability and attacking build-up, with intense activity across the midfield and into defensive areas.

The analysis of final third progressions highlights the team’s collective approach to attacking play, with both Wirtz and Xhaka featuring prominently alongside other key contributors. This data underscores Leverkusen’s emphasis on controlled, progressive play from various positions on the field.

Defensively, the spatial analysis of Tah, Tapsoba, Kossounou, and Reyna reveals a flexible structure capable of adapting between a back three and four. This tactical versatility, combined with the team’s ability to quickly transition from defense to attack, forms the foundation of Leverkusen’s success.

The passing networks and key performance indicators further illustrate the interconnectedness of Leverkusen’s play, with players like Xhaka facilitating ball movement from deep positions and Wirtz creating opportunities in advanced areas.

Overall, this analysis paints a picture of a well-balanced, tactically astute Bayer Leverkusen side. The team’s success can be attributed to its ability to leverage individual strengths within a cohesive system, blending defensive solidity with creative attacking play. As the season progresses, the continued synergy between players like Wirtz and Xhaka, supported by a robust defensive structure, will be crucial in maintaining Leverkusen’s competitive edge in both domestic and European competitions.